

Taxonomy

1. Protista group includes–

- (a) unicellular eukaryotic organisms
- (b) heterotrophic eukaryotic organisms
- (c) all green plants
- (d) None of the above

BPSC Agriculture Department-2024

Ans. (a)

The Protista group includes unicellular eukaryotic organisms. Unicellular organisms with a membrane-bound nucleus and other specialized organelles.

2. The smallest organism, capable of autonomous growth and reproduction, is

- (a) Virus
- (b) Bacteria
- (c) Mycoplasma
- (d) Bacteriophage

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Mycoplasma is the smallest known organism, capable of autonomous growth and reproduction.

3. Dolphins are classified as –

- (a) Fish
- (b) Amphibians
- (c) Reptile
- (d) Mammals

40th B.P.S.C. (Pre) 1995

Ans. (d)

Dolphins are highly intelligent aquatic mammals and are part of the family of toothed whales. They are found worldwide, mostly in shallow seas of the continental shelves and are carnivores, mostly eating fish and squid. India has declared Ganga Dolphin as the national aquatic animal.

4. Amphibia are –

- (a) Very fastly moving boats
- (b) Animals living in water only
- (c) Animals living on land only
- (d) Animals living equally in water and on land both

38th B.P.S.C. (Pre) 1992

Ans. (d)

The word 'amphibian' is derived from the ancient Greek term amphibios, which means 'both kinds of life'. The term was initially used as a general adjective for animals that could live on land or in water. They are ectotherms or cold-blooded animals, means they are unable to regulate their own body temperature.

5. The animal without red blood cells is –

- (a) frog
- (b) earthworm

- (c) snake
- (d) peacock
- (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (b)

Earthworm lack red blood cells.

6. How many lateral hearts are found in earthworm?

- (a) 6
- (b) 8
- (c) 12
- (d) More than one of the above
- (e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (b)

The Blood-Vascular system of earthworm is well developed and consists of closed blood vessels, which are longitudinal and transverse and in which red blood flows continuously. The blood is bright red in colour due to the presence of haemoglobin or erythromycin which is dissolved in plasma. Earthworm has four pair (i.e. 8) of lateral hearts.

7. Which is an atrocious animal?

- (a) Penguin
- (b) Whale
- (c) Otter
- (d) Tortoise

39th B.P.S.C. (Pre) 1994

Ans. (c)

Among the given options, otter is an atrocious animal. Otters are said to be murderous, necrophilic aqua-weasels. Otters are mammals of the subfamily Lutrinae, one of two subfamilies contained within family Mustelidae. Otters are also called serial-killing for monsters of the sea.

8. Which of the following is not true amphibian animal?

- (a) Tortoise
- (b) Frog
- (c) Toad
- (d) Salamander
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (a)

Amphibian animals that could live on land or in water. Amphibians are a diverse and exciting class of animals that include Frogs, Toads, Salamanders, newts and caecilians. So in the above given question, Tortoise is class Reptilian animal.

9. Ginger is a stem and not root because :

- (a) it stores food material
- (b) it has nodes and internodes
- (c) it grows horizontally in the soil
- (d) it lacks chlorophyll
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Stems get modified into underground structures for storage of food as seen in potato (tuber), ginger (rhizome), garlic (bulb), yam (corm). Presence of an eye (node) in potato, distinct nodes with internodes and scaly leaves in ginger/yam, a cluster of roots at the base of the reduced stem in garlic and onion, all indicate that these underground plant parts are modified stems.

10. Ginger is an underground stem and not a root because

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- (c) it lacks chlorophyll
- (d) More than one of the above
- (e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (b)

See the explanation of the above question.

11. Plants receive their nutrients mainly from –

- (a) chlorophyll
- (b) atmosphere
- (c) light
- (d) soil
- (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

Green plants are autotrophic, which means they make their organic matter (food) water and inorganic elements are obtained from the earth by plants. Inorganic elements are present in the soil in the form of minerals. These minerals are called elements or nutrients and their nutrition is called mineral nutrition.

12. When the soil is too basic, plants do not grow in it. To improve the quality, what must be added to the soil?

- (a) Organic matter
- (b) Quicklime
- (c) Calamine solution
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

When the soil is too basic, plants do not grow in it. To improve the quality organic matter (manure or compost) must be added to the soil to make it fertile.

13. The green plants capture ____ of the sunlight energy and convert it into food energy.

- (a) 5%
- (b) 2%
- (c) 1%
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

The green plants capture 1% of the sunlight energy and convert it into food energy.

14. Pea plant is :

- (a) Herb
- (b) Flower
- (c) Bush
- (d) None of these

39th B.P.S.C. (Pre) 1994

Ans. (a)

Pea plant is a dicotyledonous herbaceous annual plant. The pea plant may adapt to a variety of growing conditions, but it prefers cool damp weather with temperatures ranging from 60 to 75 degree Fahrenheit. It belongs to a large family of plants, the Fabaceae or Leguminosae.

15. Cloves are obtained from which of the following parts of the plant?

- (a) Dried leaves
- (b) Dried stems
- (c) Dried seeds
- (d) Dried flower buds

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Cloves are the aromatic dried flower buds, which is used in Ayurvedic medicines, making food and as an anodyne for dental emergencies. The botanical name of clove tree is Syzygium aromaticum. Which belongs to Myrtaceae family. The clove tree grows to a height ranging from 8-15 metres, having large oval leaves.

16. Clove, a spice, is obtained from which part of the plant?

- (a) Fruit
- (b) Stem
- (c) Root
- (d) Flower bud
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

See the explanation of the above question.

17. A plant bud is –

- (a) An embryonic shoot
- (b) An embryonic leaf
- (c) An endosperm
- (d) A seed

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

In botany, a bud is an undeveloped or embryonic shoot and normally occurs in the axil of a leaf or at the tip of the stem.

18. Among the following, which is not a true fruit?

- (a) Apple
- (b) Grape
- (c) Date
- (d) Plum
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

The true fruit can be defined as the fruit, which is formed from the fertilized ovary of the flower. Mango, grapes, cherry, watermelon, lemon, date, plum, etc. are examples of true fruits. The false fruit can be defined as the fruit, which is formed from the ripened ovary along with some other parts of the flower. Some examples of false fruit are cashew-nut, which develops from the peduncle; apple, pear, and cucumber, which develop from the thalamus; jack fruit and pineapple, which develop from the entire inflorescence, etc.

19. The part of the flower that gives rise to a fruit is

- (a) sepal (b) petal
(c) ovary (d) placenta
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

Ovary is a part of the female reproductive organ of the flower that gives rise to a fruit. Specifically, it is the part of the pistil which holds the ovule(s) and is located above or below or at the point of connection with the base of the petals and sepals.

20. Plants which grow on saline soils are-

- (a) Xerophytes (b) Hydrophytes
(c) Halophytes (d) Succulents

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Hydrophytes or aquatic plants are plants that have adapted to live in an aquatic environment. Plants growing on saline soils or water are called halophytes. They grow in saline semi-deserts, mangrove swamps, marshes and sloughs.

21. What are the characters that make a plant a hydrophyte?

- (a) Absence of mechanical tissue
(b) Presence of aerenchyma
(c) Presence of well developed vascular system
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}-2024

Ans. (d)

Hydrophytes, or aquatic plants, are adapted to live in water or in very moist conditions. These plants possess certain specialized characteristics that enable them to thrive in such environments. Key adaptations of hydrophytes include:

- **Presence of Aerenchyma**—Hydrophytes have large air-filled spaces in their tissues (aerenchyma) to facilitate buoyancy and the exchange of gases (oxygen and carbon dioxide). This helps the plant float and allows gaseous exchange even in waterlogged conditions.
- **Absence of Mechanical tissue**—Hydrophytes typically exhibit a reduced or absent mechanical tissue system compared to terrestrial plants.
- **Presence of well developed vascular system**—Hydrophytes, or aquatic plants, possess a well-developed vascular system, which is essential for their growth and survival in water-rich environments.

22. What are salt-loving plants called?

- (a) Xerophytes (b) Mesophytes

- (c) Glycophytes (d) Halophytes
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

Halophytes are the plants that thrive in water with high salinity.

23. After bringing out of water fishes die, because-

- (a) They get excess quantity of oxygen
(b) Their body temperature increases
(c) They cannot breathe
(d) They cannot move in water

39th B.P.S.C. (Pre) 1994

Ans. (c)

Fishes respire with the help of their gills. Gills can readily absorb the oxygen dissolved in water. When fishes are taken out of water, the supply of oxygen to the fishes is cut as their gills cannot absorb the oxygen present in the atmosphere. Hence, they die after sometime.

24. The interaction between algae and fungi to form lichen is called :

- (a) Parasitism (b) Mutualism
(c) Commensalism (d) Conversion
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

The interaction between algae and fungi to form lichen is called mutualism. Symbiosis of algae and fungi in lichens is the mutually helpful symbiotic relationship. The algae or cyanobacteria benefit their fungal partner by producing organic compounds through photosynthesis. In return, the fungal partner benefits the algae or cyanobacteria by protecting them from the environment by its filaments, which also gather moisture and nutrients from the environment, and (usually) provide an anchor to it.

25. In lichen, a chlorophyll-containing partner, _____ live together.

- (a) alga and a fungus (b) fungus and a bacterium
(c) alga and a virus (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

In Lichen, a chlorophyll-containing partner Alga and a fungus lives together. Lichen refers to a symbiotic association of two components, a fungus the 'mycobiont' and at least one photosynthetic organism, the 'Photobiont' consisting of a micro alga (usually a green alga) or cyanobacterium, or both.

26. Which of the following acts as a pioneer species in a xerarch succession?

- (a) Lichen
- (b) Human
- (c) Herb
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Lichen acts as a pioneer species in a xerarch succession. Pioneer species are important to the evolution of an ecosystem. The presence of barren land means the land cannot change into soil without the life processes of pioneer species. Since lichens colonize bare rock, they secrete acids that slowly break down rock. Since lichens do not need soil themselves, they are able to colonize these base areas and start the soil reaction process.

27. What is true for phylum echinodermata?

- (a) They are triploblastic and have coelomic cavity.
- (b) They have a peculiar water driven tube system.
- (c) They are exclusively free living marine animals.
- (d) More than one of the above
- (e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (d)

Echinodermata (Spiny bodied) have an endoskeleton of calcareous ossicles. All are marine with organ-system level of organisation. The adult echinoderms are radially symmetrical but larvae are bilaterally symmetrical. They are triploblastic and coelomate animals. Digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side. The most distinctive feature of echinoderms is the presence of water vascular system which helps in locomotion, capture and transport of food and respiration. An excretory system is absent. Sexes are separate. Reproduction is sexual. Fertilisation is usually external. Development is indirect with free-swimming larva.

Examples : Asterias (Star fish), Echinus (Sea urchin), Antedon (Sea lily), Cucumaria (Sea cucumber) and Ophiura (Brittle star).

28. Fungi are plants that lack –

- (a) oxygen
- (b) carbon dioxide
- (c) chlorophyll
- (d) sunlight
- (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

The Fungi does not contain chlorophyll for this reason they are no longer kept in the plant kingdom. Although earlier they were kept in the plant kingdom.

29. Which tissue increases the length of the stem and the root in plants?

- (a) Collenchyma
- (b) Apical meristem
- (c) Cambium
- (d) Parenchyma

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

Apical meristem, region of cells capable of division and growth in the root and shoot (stem) tips in plants. Apical meristems give rise to the primary plant body and are responsible for the extension of the roots and shoots.

30. Presence of Casparian strips is seen in

- (a) pericycle
- (b) endodermis
- (c) exodermis
- (d) epidermis

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

The Casparian strip is a band like thickening in the center of the root endodermis of vascular plants (Pteridophytes and Spermatophytes). Presence of Casparian strip is seen in endodermis.

31. Which plant is not the example of algae?

- (a) Fern
- (b) Spirogyra
- (c) Chara
- (d) Ulothrix

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (a)

Ferns are plants that do not have flowers. Ferns generally reproduce by producing spores. Similar to flowering plants, ferns have roots, stems and leaves. So, Fern is not the example of algae.

32. Which of the following organisms has no blood but respire?

- (a) Cockroach
- (b) Fish
- (c) Hydra
- (d) Earthworm
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

Hydra are truly fascinating small aquatic animals. They can be found in fresh water ponds and slow-moving rivers, where they usually attach themselves to plants or rocks. Hydra does not have blood. In the hydra, respiration is accomplished via diffusion of molecules of oxygen and carbon dioxide through the skin whereas cockroach, fish and earthworm have blood and blood help in their respiration.

33. Which of the following is an example of the binomial scientific name?

- (a) Green algae
- (b) Snow leopard